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DS 4002: CS3 Hook Document
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The Night Shift Model: Classifying Chest X-rays When Every Minute Counts Scenario

It's 2:14 AM at the University of Virginia Hospital. A large wave of patients just arrived in the emergency department, and the radiology team is overwhelmed. A stack of chest X-rays sits unreviewed, and some may show signs of serious conditions like pneumonia or collapsed lungs, but no one knows which ones yet. A physician turns to you, the data scientist on call, and says:

“We need to know which scans look abnormal. Can your model help us triage faster?”

As the Data Scientist on call at the hospital, you bridge the gap between complex deep-learning models and real-world medical decisions, ensuring that the tools you build are fast, reliable, and trustworthy when the stakes are highest. At this moment, you're not a student completing an assignment, but rather you're part of the care team. Your job is to build a tool that helps clinicians focus their attention where it's needed most.

Your Task

In this case study, you'll step into the role of a machine learning engineer responsible for building a system that can distinguish normal from abnormal medical images. You will be asked to prepare image data, design a classification approach, and evaluate whether your model is reliable enough to support clinical decision-making.

Your mission is to create a clear, functional demonstration of how deep learning can be used to flag potentially abnormal scans. You'll decide how to structure your pipeline, how to train and test your model, and how to communicate your results in a way that a clinician or stakeholder could understand. The exact requirements and deliverables will be detailed in the rubric.

Why this Matters

Chest X-rays are one of the most common medical imaging tools in the world, yet many hospitals face shortages of radiologists and long review times. Automated systems that flag potentially abnormal scans can help clinicians prioritize urgent cases and reduce delays in care. This is your chance to build something that feels real, relevant, and impactful.

Welcome to the night shift. Let's see what your model can do!

Link to Github repository: <https://github.com/anutishah/DS4002-CS3>